

Dr. Félix P. Leiva

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Radboud University
Nijmegen, Netherlands
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EDUCATION

- 2022 **Doctorate** – Radboud University, The Netherlands. Thermal biology and cell size: an oxygen limitation perspective in ectotherms.
- 2011 **Aquaculture Engineer** – Universidad Austral de Chile, Chile. Respiratory physiology of the ghost shrimp *Neotrypaea uncinata* (Decapoda: Thalassinidea) during hypoxia and reoxygenation.
- 2007 **Bachelor in Science and Technology** – Universidad Austral de Chile, Chile.

APPOINTMENTS

- 2025 – present **Postdoctoral position** – Radboud University, The Netherlands. Project: Power of Size.
- 2023 – 2024 **Alexander von Humboldt fellow** – Alfred Wegener Institute for Polar and Marine Research, Germany. Project: Metabolism of life on Earth across aquatic and terrestrial realms.
- 2022 – 2022 **Postdoctoral position** – Radboud University, The Netherlands. Project: ShareTrait, a data portal for making trait data interoperable and reusable.
- 2017 – 2022 **PhD position** – Radboud University, The Netherlands. Project: From cells to life-histories: explaining the slow-fast life-style continuum in ectotherms from thermal constraints on resource acquisition.
- 2013 – 2017 **Research assistant** – Centro i~mar, Universidad de Los Lagos, Chile. Project: Diversity of life-histories, connectivity and variability in the relative contribution of estuarine and coastal habitats to the SE Pacific population of Southern hake *Merluccius australis*.
- 2014 – 2015 **Research assistant** – Centro i~mar, Universidad de Los Lagos, Chile. Project: Origen natal y distribución geográfica de la reineta en Chile/Natal origen and geographic distribution of the pacific pomfret in Chile.
- 2013 – 2014 **Research assistant** – Centro i~mar, Universidad de Los Lagos, Chile. Project: Rol ecosistémico de la sardina austral e impacto de su explotación sobre la sustentabilidad de otras especies de interés comercial/Ecosystemic role of

Patagonian sprat and impact of their exploitation on the sustainability of other species of commercial interest.

- 2013 **Research assistant** – Laboratorio de Investigación Pesquera, Universidad Austral de Chile, Chile. Project: Population dynamics of *Mytilus chilensis* in the Reloncaví Fjord: Scientific bases for mussel aquaculture.
- 2011 **Research assistant** – Laboratorio de Ecofisiología de Crustáceos LECOFIC, Universidad Austral de Chile, Chile. Project: Posicionamiento del cultivo de centolla austral para la diversificación de la acuicultura de especies nativas/Positioning the Southern king crab farming for diversification aquaculture.
- 2007 – 2009 **Ad honorem** technician – Laboratorio de Ecofisiología de Crustáceos LECOFIC, Universidad Austral de Chile, Chile.
- 2006 **Ad honorem** technician – Hatchery de Invertebrados Marinos, Universidad Austral de Chile, Chile.

NON-ACADEMIC EXPERIENCE

- 2000 – 2012 **Artisanal fisherman** – Fishery of Southern hake, pink cusk-eel and ray in Reloncaví Sound and Ancud Gulf during summer and winter “holidays” **over 12 years.**

ACADEMIC AWARDS AND FUNDING (TOTAL FUNDED: ~159,000 EUR)

- 2025 **Company of Biologists Travel Grant**, The Company of Biologists, to attend the “Society for Experimental Biology Annual Conference” held in Antwerp, Belgium, from July 07 to July 11, 2025. ~420 EUR.
- 2023 **SEB Conference Travel Grant**, The Society for Experimental Biology, to attend the “SEB Annual Conference” held in Edinburgh, Scotland, from July 4 to July 7, 2023. ~680 EUR.
- 2022 **Humboldt Research Fellowship for Postdocs**, Alexander von Humboldt Foundation, to pursue postdoctoral research at the Alfred Wegener Institute (AWI), Bremerhaven, Germany. ~83,000 EUR over two years.
- 2022 **Company of Biologists Travel Grant**, The Company of Biologists, to attend the “14th International Congress on the Biology of Fish (ICBF)” held in Montpellier, France, from June 28 to July 1, 2022. ~580 EUR.
- 2018 **Becas Chile scholarship**, National Agency for Research and Development (ANID), Chile. 90,000 USD (~74,000 EUR) over two and a half years.

- 2011 **Artisanal fisherman son's scholarship** for undergraduate studies (Aquaculture Engineering) by Fundación Chinchihue, Chile. 300,000 CLP (~345 EUR).
- 2009 **Artisanal fisherman son's scholarship** for undergraduate studies (Aquaculture Engineering) by Consejo Regional de Pescadores Artesanales (COREPA), Chile. 600,000 CLP (~690 EUR).

PUBLICATIONS (ORCID ID: 0000-0003-0249-9274; *CORRESPONDING AUTHOR; **STUDENT MENTEE) [link to Google Scholar]

1. **Leiva FP***, Molina-Venegas R, Alter K, Freire CA, Hendriks AJ, Hermaniuk A, Serre-Fredj L, Shokri M, Czarnoleski M, Mark FC. 2026. ErythroCite: a database on red blood cell size of fishes. Scientific Data. <https://doi.org/10.1038/s41597-026-06904-1>. [link to paper].
2. Verberk WCEP, Henry E, **Leiva FP**, Barbarossa V, Schipper A. 2026. Heat tolerance and its plasticity in freshwater and marine fishes reflect exposure to extremes and seasonal variation in habitat temperatures. Heat tolerance and its plasticity in freshwater and marine fishes reflect exposure to extremes and seasonal variation in habitat temperatures. Ecology Letters 29, e70341. <https://doi.org/10.1111/ele.70341>. [link to paper].
3. Kempf S, Neven CJ, Timpe A, **Leiva FP**, Seibel B, Mark FC. 2025. The physiological response of the Arctic key species Polar cod, *Boreogadus saida*, to hypoxia in a warming ocean: critical oxygen levels and swimming performance. <https://doi.org/10.1186/s40850-025-00241-3>. BMC Zoology 10, 1-12. [link to paper].
4. **Leiva FP***, Ellers J, Berg MP, Cuxart-Erruz R, Barneche D, Blackburn T, Castañeda LE, Chown S, Gaitán-Espitia JD, Gebauer P, Gomez Isaza D, Hardy I, Hermaniuk A, Hirst A, Jorissen S, Keasar T, Koene JM, Le Lann C, Martorelli I, Molinet C, Niklitschek EJ, Oliveira B, Olivier B, Orizaola G, Pilakouta N, Shameer KS, Shokri M, Stoks R, Tougeron K, Tuni C, van de Pol ILE, van Dis NE, Visser B, Vogels J, White CR, Wu N, Verberk WCEP. (2025). ShareTrait: towards interoperable and reusable individual trait-based data in ectotherms. Functional Ecology 39, 3124-3138. <https://doi.org/10.1111/1365-2435.70147>. [link to paper].
5. Babin CH, **Leiva FP**, Verberk WCEP, Rees BB. 2024. Evolution of key oxygen-sensing genes is associated with hypoxia tolerance in fishes. Genome Biology and Evolution 16, evae183. <https://doi.org/10.1093/gbe/evae183>. [link to paper].
6. **Leiva FP***, Santos M, Rezende EL, Verberk WCEP. 2024. Intraspecific variation of heat tolerance in a model ectotherm: The role of oxygen, cell size and body size. Functional Ecology 38, 439-448. <https://doi.org/10.1111/1365-2435.14485>. [link to paper].
7. **Leiva FP***, Boerrigter JGJ, Verberk WCEP. 2023. The role of cell size in shaping responses to oxygen and temperature in fruit flies. Functional Ecology 37, 1269-1279. <https://doi.org/10.1111/1365-2435.14294>. [link to paper]. Shortlisted for the 2023 Haldane Prize.

8. Verberk WCEP*, Sandker JF, van de Pol I, Urbina M, Wilson R, McKenzie DJ, **Leiva FP**. 2022. Body mass and cell size shape the tolerance of fishes to low oxygen in a temperature-dependent manner. *Global Change Biology* 28, 5695-5707. <https://doi.org/10.1111/gcb.16319>. [link to paper].
9. Verspagen N**, **Leiva FP***, Janssen IM, Verberk WCEP. 2020. Effects of developmental plasticity on heat tolerance may be mediated by changes in cell size in *Drosophila melanogaster* may be mediated by changes in cell size. *Insect Science* 27, 1244-1256. <https://doi.org/10.1111/1744-7917.12742>. [link to paper].
10. Toledo P*, Niklitschek EJ, Darnaude AM, **Leiva FP**, Harrod C, Lillo S, Ojeda V, Klarian S, Molina-Burgos B, Gálvez P, Canales-Aguirre CB. 2020. The trophic ecology of partial migration: insights from *Merluccius australis* off NW Patagonia. *ICES Journal of Marine Science* 77, 1927-1940. <https://doi.org/10.1093/icesjms/fsaa065>. [link to paper].
11. **Leiva FP***, Calosi P, Verberk WCEP. 2019. Scaling of thermal tolerance with body mass and genome size in ectotherms: A comparison between water-and air-breathers. *Philosophical Transactions of the Royal Society B Biological Sciences* 374, 20190035. <https://doi.org/10.1098/rstb.2019.0035>. [link to paper].
12. Sunday J*, Bennett JM, Calosi P, Clusella-Trullas S, Gravel S, Hargreaves AL, **Leiva FP**, Verberk WCEP, Olalla-Tárraga MA, Morales-Castilla I. 2019. Thermal tolerance patterns across latitude and elevation. *Philosophical Transactions of the Royal Society B Biological Sciences* 374, 20190036. <https://doi.org/10.1098/rstb.2019.0036>. [link to paper].
13. Toledo P*, Darnaude AM, Niklitschek EJ, Ojeda V, Voue R, **Leiva FP**, Labonne M, Canales-Aguirre CB. 2019. Partial migration and early growth of southern hake *Merluccius australis*: a journey between estuarine and oceanic habitats off Northwest Patagonia. *ICES Journal of Marine Science* 76, 1094-1106. <https://doi.org/10.1093/icesjms/fsy170>. [link to paper].
14. Murúa P*, Patiño DJ, **Leiva FP**, Küpper FC, Müller DG, Peters AF, Muñoz LA, Westermeier R. 2019. Gall disease in the alginophyte *Lessonia berteroana*: a pathogenic interaction linked with host adulthood in a seasonal-dependant manner. *Algal Research* 39, 101435. <https://doi.org/10.1016/j.algal.2019.101435>. [link to paper].
15. **Leiva FP***, Garcés C, Verberk WCEP, Care M, Paschke K, Gebauer P. 2018. Differences in the respiratory response to temperature and hypoxia across four life-stages of the intertidal porcelain crab *Petrolisthes laevis*. *Marine Biology* 165, 146-156. <https://doi.org/10.1007/s00227-018-3406-z>. [link to paper].
16. **Leiva FP***, Niklitschek EJ, Gebauer P, Paschke K, Urbina MA. 2016. Tide-related biological rhythm in the oxygen consumption rate of ghost shrimp (*Neotrypaea uncinata* Milne Edwards). *The Journal of Experimental Biology* 219, 1957-1960. <https://doi.org/10.1242/jeb.133785>. [link to paper].

17. **Leiva FP**, Urbina MA, Cumillaf JP, Gebauer P, Paschke K*. 2015. Physiological responses of *Neotrypaea uncinata* (Milne Edwards 1837) (Decapoda: Thalassinidea) to oxygen availability and recovery after severe environmental hypoxia. *Comparative Biochemistry and Physiology-Part A Molecular and Integrative Physiology* 189, 30-37. <https://doi.org/10.1016/j.cbpa.2015.07.008>. [link to paper].

MANUSCRIPTS IN PREPRINT

Leiva FP, Verberk WCEP, Calosi P, Rezende EL, Mark FC. 2024. MetaR, a global database on metabolic rates of ectotherms. *EcoEvoRxiv*. <https://doi.org/10.32942/X2Z022>. [link to preprint].

Leiva FP, Santos M, Niklitschek EJE, Rezende EL, Verberk WCEP. 2024. Genetic variation of heat tolerance in a model ectotherm: an approach using thermal death time curves. *EcoEvoRxiv*. <https://doi.org/10.32942/X20C9T>. [link to preprint].

DATASETS AND CODE [link to GitHub]

Babin CH, **Leiva FP**, Verberk WCEP, Rees BB. 2023. Paper data and code of the manuscript: Evolution of oxygen-sensing genes parallels hypoxia tolerance in fishes. Zenodo. <https://doi.org/10.5281/zenodo.10026472>.

Leiva, FP, Ellers J, Berg MP, Cuxart-Erruz R, Barneche D, Blackburn T, Castañeda LE, Chown S, Gaitán-Espitia JD, Gebauer, P, Gomez Isaza D, Hardy I, Hermaniuk A, Hirst A, Jorissen S, Keasar T, Koene JM, Le Lann C, Martorelli I, Molinet C, Niklitschek EJE, Oliveira B, Olivier B, Orizaola G, Pilakouta N, Shameer KS, Shokri M, Stoks R, Tougeron K, Tuni C, van de Pol ILE, van Dis NE, Visser B, Vogels J, White CR, Wu N, and Verberk WCEP. (2025). ShareTrait: a data portal for making trait data interoperable and reusable (version 1.2.0) [Data set]. Zenodo. <https://doi.org/10.5281/zenodo.14826294>.

Leiva FP, Boerrigter JGJ, Verberk WCEP. 2023. Paper data and code of manuscript: The role of cell size in shaping responses to oxygen and temperature in fruit flies. Zenodo. <https://doi.org/10.5281/zenodo.5949048>.

Verberk WCEP, Sandker JF, van del Pol I, Urbina M, Wilson R, McKenzie DJ, **Leiva FP**. 2022. Paper data and code of manuscript: Body mass and cell size shape the tolerance of fishes to low oxygen in a temperature-dependent manner. Zenodo. <https://doi.org/10.5281/zenodo.6123770>.

Leiva FP, Santos M, Rezende E, Verberk WCEP. 2021. Paper data and code of manuscript: Intraspecific variation on heat tolerance in a model ectotherm: effects of body mass, cell size, oxygen and sex. Zenodo. <https://doi.org/10.5281/zenodo.5120028>.

Verspagen N, **Leiva FP**, Janssen IM, Verberk WCEP. 2020. Data and code from: Effects of developmental plasticity on heat tolerance may be mediated by changes in cell size in *Drosophila melanogaster*. DANS. <https://doi.org/10.17026/dans-z2a-9m4g>.

Sunday J, Bennett JM, Calosi P, Clusella-Trullas S, Gravel S, Hargreaves AL, **Leiva FP**, Verberk WCEP, Olalla-Tárraga MA, Morales-Castilla I. 2019. Data from: Thermal tolerance patterns across latitude and elevation. Dryad Digital Repository. <https://doi.org/10.5061/dryad.5002200>.

Leiva FP, Calosi P, Verberk WCEP. 2019. Data from: Scaling of thermal tolerance with body mass and genome size in ectotherms: a comparison between water- and air-breathers. Dryad Digital Repository. <http://doi.org/10.5061/dryad.878vn25>.

Leiva FP, Garcés C, Verberk WCEP, Care M, Paschke K, Gebauer P. 2018. Data and code of manuscript: Differences in the respiratory response to temperature and hypoxia across four life-stages of the intertidal porcelain crab *Petrolisthes laevis* (Version 1.0). Zenodo. <http://doi.org/10.5281/zenodo.5120850>.

Leiva FP. 2011. Fisiología respiratoria del camarón fantasma *Neotrypaea uncinata* (Milne Edwards, 1837) (Decapoda: Thalassinidea) durante la hipoxia y reoxigenación. Zenodo. <https://doi.org/10.5281/zenodo.4738962>.

TEACHING OVERVIEW

2025 – present	Coordinator, Radboud University, the Netherlands. MSc Course: Orientation in Biology and Environmental Sciences (over 50 students).
2022	Guest lecturer (<i>ad honorem</i>), Radboud University, the Netherlands. MSc Course: Adaptation Physiology (20 students).
2020	Guest lecturer (<i>ad honorem</i>) and tutor (on systematic review, R, and reproducibility), Universidad de Los Lagos, Chile. PhD Course: Life histories of marine invertebrates (6 students).
2009 – 2010	Teaching assistant, Universidad Austral de Chile, Chile. Course: Animal Biology (over 30 students).

STUDENT SUPERVISION (2 MASTER AND 2 BACHELOR STUDENTS)

2026	Master student. Topic: A systematic mapping and meta-analysis of the impacts of chemicals on animal organ sizes.
2019 – 2020	Master student. Topic: Drivers of heat tolerance in fish: the role of phylogeny, body size, genome size, geographic origin, migration, and habitat.
2020	Bachelor student. Topic: Growth rates in crustaceans: disentangling effects of body size, genome size and temperature.
2019	Bachelor student. Topic: The consequences of cell size on the sensitivity for oxygen.

2017 – 2018 **Master student.** Topic: The influence of cell size on thermal tolerance in *Drosophila melanogaster*.

RELEVANT SKILLS

Computational	Experience as a user of Windows and Linux (Ubuntu, Debian). Experience in statistical modelling (e.g., PGLS, LME, GLMM) and Bayesian inference using R (intermediate user). Experience in database design, creation and data curation using Microsoft Access and LibreOffice Base. Experience in version control platforms (GitHub).
Field	Experience in research cruises both in artisanal fishery boats and scientific vessels (until 25 days onboard). Installation and operation of several mid-water trawls (e.g., IKMT). Manufacture, reparation and operate artisanal fishing gear such as longlines and gill-net.
Laboratory	Respirometry measurements on various species of decapod crustaceans. Spectrophotometric analysis of blood metabolites after exposure to extreme environmental conditions (oxygen, temperature, salinity) in the laboratory. Heat tolerance trials using static approaches (thermal death time curves). Sampling and preparation of seawater, phytoplankton, animal and seaweed tissues for stable isotope analysis. Mounting, sectioning, polishing, and micro-drilling of different fish otoliths, both for age estimation and for stable isotopes and metals (LA-ICP-MS, ICP-MS) analysis. Rearing of southern king crab (<i>Lithodes santolla</i>) and fruit fly (<i>Drosophila melanogaster</i>).

CONFERENCES

Leiva FP, Molina-Venegas R, Alter K, Freire CA, Hendriks AJ, Hermaniuk A, Serre-Fredj L, Shokri M, Czarnoleski M, Mark FC. 2025. ErythroCite: a systematic map and open-source database on red blood cell size of fishes. Society for Experimental Biology Annual Meeting. Antwerp, Belgium (Poster).

Shokri M, Potapov A, De Leo F, **Leiva FP**, Uvarov LV, Basset A. Metabolic responses to climate change in invertebrates. Society for Experimental Biology Annual Meeting. Antwerp, Belgium (Poster).

Leiva FP, Verberk WCEP, Calosi P, Rezende EL, Mark FC. 2024. MetaR, a global database on metabolic rates of ectotherms. Society for Experimental Biology Annual Meeting. Prague, Czech Republic (Talk).

Babin CH, **Leiva FP**, Verberk WCEP, Rees BB. 2024. Evolution of oxygen-sensing genes parallels whole-animal hypoxia tolerance in fishes. The Society for Integrative and Comparative Biology (SICB), Seattle, USA (Poster).

Darnaude A, Labonne M, **Leiva FP**, Panfili J, Ferraton F, Niklitschek EJ. 2023. Diversity in larval origin, nursery habitat suitability and early growth in the southern hake *Merluccius australis* in NW Patagonia. 7th International Otolith Symposium, Chile (Poster).

Leiva FP, Santos M, Rezende E, Verberk WCEP. 2023. Variation of thermal death time curves in a model ectotherm. Society for Experimental Biology Annual Meeting. Edinburgh, Scotland (Talk).

Verberk WCEP, **Leiva FP**, Ellers J, Berg MP. 2022. ShareTrait: a data portal for making trait data interoperable and reusable. British Ecological Society Annual Meeting. Edinburgh, Scotland (Poster).

Leiva FP, Verberk WCEP, Mark FC. 2022. Metabolism of life on Earth across aquatic and terrestrial realms. Society of Experimental Biology Animal Satellite Meeting “Why and how does metabolic rate scale with body size?”. Montpellier, France (Poster).

Verberk WCEP, Sandker JF, van de Pol I, Urbina MA, Wilson RD, McKenzie DJ, **Leiva FP**. 2022. Body mass and genome size shape the tolerance of fishes to low oxygen in a temperature-dependent manner. 14th International Congress on the Biology of Fish (ICBF). Montpellier, France (Talk).

Leiva FP, Boerrigter JGJ, Verberk WCEP. 2022. The role of cell size in shaping responses to oxygen and temperature in fruit flies. Society of Experimental Biology Annual Meeting. Montpellier, France (Talk).

Leiva FP, Calosi P, Verberk WCEP. 2021. Scaling of thermal tolerance with body mass and genome size in ectotherms: A comparison between water-and air-breathers. Society of Experimental Biology Annual Meeting. Online (Talk).

Toledo P, Darnaude A, Niklitschek EJ, Ojeda V, Voué R, **Leiva FP**, Labonne M, Canales-Aguirre C. 2019. Partial migration and early growth of southern hake *Merluccius australis*: a journey between estuarine and oceanic habitats off Northwest Patagonia. XII Meeting of the Chilean Association of Ichthyology, Puerto Montt, Chile (Talk).

Niklitschek EJ, Toledo P, **Leiva FP**, Garcés C. 2019. Isotopic landscapes, environmental variability and structure of fish populations in South America and Antarctica: insights from otolith chemistry. XII Meeting of the Chilean Association of Ichthyology, Puerto Montt, Chile (Talk).

Leiva FP, Garcés C, Verberk WCEP, Care M, Paschke K, Gebauer P. 2017. Differences in the respiratory response to temperature and hypoxia across four life stages of the intertidal porcelain crab *Petrolisthes laevis*. Ecology Across Borders: Joint Annual Meeting, Ghent, Belgium (poster).

Garcés C, **Leiva FP**, Toledo P, Neira S, Niklitschek EJ. 2017. Trophic ecology of native predator fish and Chinook salmon in the Inner Sea of Chiloé. XXXVII Congress on Marine Sciences, Valparaiso, Chile (poster).

Niklitschek EJ, Toledo P, Garcés C, Neira S, **Leiva FP**. 2016. Trophic roles of commercially important fishes in the Inner Sea of Chiloé, as revealed by stable isotope analysis. XXXVI Congress on Marine Sciences, Concepción, Chile (talk).

Garcés C, **Leiva FP**, Lillo S, Neira S, Niklitschek EJ. 2016. Trophic interaction between Southern hake, Patagonian grenadier and Patagonian sprat in the Inner Sea of Chiloé. XXXVI Congress on Marine Sciences, Concepción, Chile (talk).

Niklitschek EJ, Dagneau de Richencour D, **Leiva FP**, Ferraton F, Petit C, Darnaude AM, Toledo P. 2016. Identifying the main nursery areas of Southern hake *Merluccius australis* in the northern Patagonian Fjords System and adjacent oceanic waters. XXXVI Congress on Marine Sciences, Concepción, Chile (talk).

Leiva FP, Niklitschek EJ, Paschke K, Gebauer P, Urbina MA. 2016. Metabolic synchrony in the ghost shrimp (*Neotrypaea uncinata* Milne Edwards): Evidence of tide-related biological rhythm. XXXVI Congress on Marine Sciences, Concepción, Chile (poster).

Leiva FP, Urbina MA, Cumillaf JP, Gebauer P, Paschke K. 2015. Physiological responses of *Neotrypaea uncinata* to oxygen availability and recovery after severe environmental hypoxia. XXXV Congress on Marine Sciences, Coquimbo, Chile (talk).

Leiva FP, Balboa C, Paschke K. 2010. Embryonic respiration of *Neotrypaea uncinata* (Decapoda: Thalassinidea) in function of pressure partial of oxygen. XXX Congress on Marine Sciences, Concepción, Chile (poster).

INVITED PEER REVIEWER FOR:

Journal Experimental Marine Biology and Ecology (1), Marine Ecology Progress Series (1), Nature Climate Change (1), Proceedings of the Royal Society B: Biological Sciences (2), The Biological Bulletin (1), Oikos (1), Biology Open (3), Journal of Thermal Biology (1).

PROFESSIONAL SOCIETY MEMBERSHIPS

Society for Open, Reliable, and Transparent Ecology and Evolutionary Biology (SORTEE; member since 2020).

The Society for Experimental Biology (SEB; member since 2017).

REFEREES

Researcher Dr **Felix C. Mark**

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Relationship: Postdoctoral mentor

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Relationship: Mentor

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Relationship: PhD supervisor